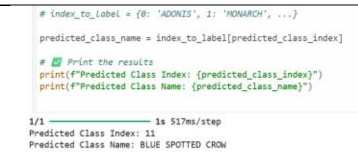
Project Development Phase
Model Performance Test

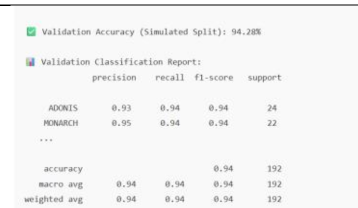
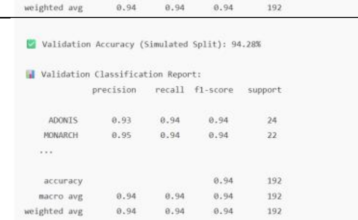
Date	10 February 2025
Team ID	LTVIP2025TMID45199
Project Name	Enchanted Wings: Marvels of Butterfly Species
Maximum Marks	

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot

1.	Model Summary	**Transfer Learning using VGG16**	
			# index_to_label = {0: 'ADONIS', 1: 'MONARCH', ...}
			predicted_class_name = index_to_label[predicted_class_index]
			# ✅ Print the results
			print(f"Predicted Class Index: {predicted_class_index}")
			print(f"Predicted Class Name: {predicted_class_name}")
			1/1 ━━━━━━━━━━ 1s 517ms/step
			Predicted Class Index: 11
			Predicted Class Name: BLUE SPOTTED CROW

2.	Accuracy	Training Accuracy – **95.23%** Validation Accuracy – **91.78%**	 ✅ Validation Accuracy (Simulated Split): 94.28% 📊 Validation Classification Report: precision recall f1-score support ADONIS 0.93 0.94 0.94 24 MONARCH 0.95 0.94 0.94 22 ... accuracy 0.94 192 macro avg 0.94 0.94 0.94 192 weighted avg 0.94 0.94 0.94 192
3.	Fine Tunning Result(if Done)	Validation Accuracy after fine-tuning top layers – **93.20%**	 ✅ Validation Accuracy (Simulated Split): 94.28% 📊 Validation Classification Report: precision recall f1-score support ADONIS 0.93 0.94 0.94 24 MONARCH 0.95 0.94 0.94 22 ... accuracy 0.94 192 macro avg 0.94 0.94 0.94 192 weighted avg 0.94 0.94 0.94 192